

Question 6: Explain the role of Machine Learning in disease prediction and early diagnosis.

Related SDG: SDG 3 – Good Health and Well-being

Problem Statement

Early detection of diseases is important for improving patient health and reducing treatment costs. Machine Learning (ML) can analyze medical records and predict diseases before symptoms become severe.

Background

Hospitals generate a large amount of patient data, including age, blood pressure, glucose level, cholesterol, and medical history. ML algorithms learn patterns from this data and help doctors make accurate decisions.

Learning Problem

- **Input Features:** Age, blood pressure, glucose level, cholesterol, BMI, medical history.
- **Target Output:** Disease prediction (Positive/Negative).

Machine Learning Approach

Decision Tree is suitable because it is simple, accurate, and easy to interpret by doctors.

Workflow

1. Collect patient data.
2. Clean and preprocess the data.
3. Split data into training and testing sets.
4. Train the Decision Tree model.
5. Predict disease status.
6. Evaluate accuracy using performance metrics.

Expected Outcome

- Early disease detection.
- Improved diagnosis accuracy.
- Faster medical decisions.
- Reduced healthcare costs.

Advantages

- Fast predictions.
- High accuracy.
- Easy to understand.
- Supports doctors in decision-making.

Limitations

- Requires quality medical data.
- Accuracy depends on dataset size.
- Cannot completely replace doctors.

Practical Applications

- Diabetes prediction
- Heart disease prediction
- Cancer diagnosis
- Kidney disease detection

Conclusion

Machine Learning improves healthcare by predicting diseases at an early stage, helping doctors provide timely treatment and supporting **SDG 3: Good Health and Well-being**.

Question 24: Explain traffic congestion prediction using Machine Learning.

Related SDG: SDG 11 – Sustainable Cities and Communities

Problem Statement

Traffic congestion causes delays, pollution, and fuel wastage. Machine Learning can predict traffic conditions and help improve traffic management.

Background

Traffic data is collected from GPS devices, road sensors, cameras, and weather reports. ML models analyze this data to forecast congestion.

Learning Problem

- Input Features: Vehicle count, road speed, weather, time, day, accidents.
- Target Output: Traffic level (Low, Medium, High).

Machine Learning Approach

A Decision Tree model can classify traffic conditions based on different road and environmental factors.

Workflow

1. Collect traffic data.
2. Clean and preprocess the dataset.
3. Train the Decision Tree model.
4. Predict future traffic conditions.
5. Display congestion alerts.

Expected Outcome

- Reduced traffic congestion.
- Better route planning.
- Lower fuel consumption.

- Improved emergency response.

Advantages

- Real-time traffic prediction.
- Reduces travel time.
- Supports smart city planning.
- Improves road safety.

Limitations

- Requires continuous data collection.
- Accuracy decreases with incomplete data.
- Needs regular model updates.

Practical Applications

- Google Maps traffic prediction
- Smart traffic signals
- Navigation systems
- Urban transportation planning

Conclusion

Machine Learning helps cities predict traffic congestion and manage transportation efficiently, contributing to SDG 11: Sustainable Cities and Communities.